

# LucyTrace

## Turn "Ruff Drafts" into the "Director's Cut"

**LucyTrace** is a high-performance pipeline for converting raster images (PNG, JPG, WEBP, etc.) into clean vector SVGs and print-ready coloring pages.

Named after our Director of Wellness (Lucy the Cavachon), this tool ensures your images are well-groomed, free of noise, and ready for:

1. **Laser Cutting/Engraving:** Compatible with LightBurn, XTool Creative Space, and Glowforge.
2. **CAD/CNC:** Optimized node counts for smoother import into **Autodesk Fusion** sketches.
3. **Print:** Generates high-resolution, centered PDFs and PNGs for coloring books.

## Quick Start

1. **Drop your images** (PNG, JPG, BMP, etc.) into the root folder next to the script.
2. **Run the pipeline:**
  - **Windows:** Double-click `lucytrace.bat`.
  - **Mac/Linux:** Double-click `lucytrace.sh`.
3. **Choose your Profile** from the menu:
  - `[1] BASE` : Fast processing for clean line art.
  - `[2] ADVANCED` : Smart processing for photos, sketches, or complex images.
4. **Collect your files:**
  - `output_svg/` -> Vector files.
  - `output_png/` -> High-res raster files.
  - `output_pdf/` -> Print-ready documents.

## The Two Profiles

LucyTrace uses a unified engine but offers two distinct "grooming" styles:

### 1. Base Profile ("The Fast Track")

- **Best For:** High-quality scans, crisp digital line art, black ink on white paper.
- **Logic:** Applies a strict brightness threshold (default 60%). Pixels darker than the limit become black; everything else becomes white.
- **Pros:** Extremely fast and predictable on clean input.

### 2. Advanced Profile ("The Director's Cut")

- **Best For:** Photos of drawings (bad lighting), "muddy" scans, comic book panels, or inverted images (white lines on black).
- **Logic:**
  - **Auto-Levels:** Stretches contrast to fix lighting issues (makes dark grays black, light grays white).
  - **Smart Detection:** Automatically detects if an image is Color or B&W.
  - **Posterization:** If color is detected, it flattens gradients into chunks ("stained glass" style) before tracing to prevent noise.

## Configuration ( `lucytrace.ini` )

You don't need to edit the Python code to change settings. All major toggles are in `lucytrace.ini` :

```
[General]
profile = adv          ; Default profile if running via CLI
overwrite = false     ; Set to true to force reprocessing existing files
export_width = 3000    ; Width of the final PNG (3000px is approx 10 inches at 300 DPI)

[ImageProcessing]
threshold_percent = 60% ; The cutoff for black vs white

[Modes]
invert = off           ; Set to 'on' if your input is white lines on a black background
mode = auto            ; Can force 'color' or 'bw' manually here
```

## 📁 Folder Structure

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The project uses the following structure (folders are auto-created if missing):

```
<project folder>/
├─ input_images/    # Source images (auto-moved here from root)
├─ cleaned_png/     # Intermediate black/white PNGs (useful for debugging)
├─ output_svg/      # Traced vector output files (Optimized for CAD/Laser)
├─ output_png/      # Exported PNGs for printing
├─ output_pdf/      # Exported PDFs for printing
├─
├─ lucytrace.py     # The Master Script
├─ lucytrace.ini    # Configuration File
├─ lucytrace.bat    # Windows Launcher
├─ lucytrace.sh     # Mac/Linux Launcher
└─ README.md
```

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## 📋 Requirements

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All tools must be installed and accessible via your system PATH.

### 1. Python 3.x

- **All OS:** Download from [Python.org](https://python.org).

### 2. ImageMagick (for cleaning)

- **Windows:** Download the **DLL version** from [ImageMagick.org](https://imagemagick.org).
  - **IMPORTANT:** During installation, check "**Install legacy utilities (e.g. convert)**" and "**Add to PATH**".
- **Mac/Linux:** `brew install imagemagick` or `sudo apt install imagemagick`.

### 3. Potrace (for vectorizing)

- **Windows:** Download the binary from [SourceForge](https://sourceforge.net/projects/potrace). Add the folder containing `potrace.exe` to your System PATH.
- **Mac/Linux:** `brew install potrace` or `sudo apt install potrace`.

### 4. Inkscape (for exporting)

- **Windows:** Download and install from [Inkscape.org](https://inkscape.org).
- **Mac/Linux:** `brew install --cask inkscape` or `sudo apt install inkscape`.

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*Happy Making!* 🎉